



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO S.P.MOHAN & CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.17 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.378384, 78.162582)



Welcome to the Journey Risk Management Study

To provide a comprehensive route safety analysis for the journey from CVQF+23W, Sangagiri, Tamil Nadu to 95H7+83Q, Tamil Nadu, covering approximately 44.17 kilometers, let's examine the various points outlined:

1. Overview of the Route Map:

- The route begins in Sangagiri and typically traverses through a mix of rural and semi-urban areas. It involves traveling primarily on National and State Highways, which are generally well maintained and facilitate heavy vehicle transit.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with significant variations. During the monsoon season (October to December), heavy rainfall can lead to waterlogging and slippery roads. Summers can be extremely hot, which might affect vehicles' tires and cooling systems.

3. Traffic Patterns:

- Peak hours typically occur early in the morning (7-9 AM) and late afternoon (5-7 PM). Congestion-prone areas include market zones near Sangagiri and major intersections on State Highway routes.

4. Assessment of Road Quality and Infrastructure:

- Roads are generally in good condition but watch for occasional potholes, especially post-monsoon season. Highways are often maintained well, but rural connecting roads may need caution due to uneven surfaces.

5. Suggestions for Alternative Routes:

- In case of emergencies, consider state highway diversions available en route. Local roads connecting small towns may serve as feasible alternatives, albeit with slower progress due to lesser infrastructure.

6. Local Regulations Affecting Hazardous Material Transport:

- Ensure compliance with local laws requiring permits for hazardous material transport. It is essential to follow time restrictions, especially during peak hours in populated areas to minimize risk.

7. Historical Incidents:

- There have been few incidents involving heavy vehicles in this region. Notably, mechanical failures and driver fatigue have been common factors in past accidents.

8. Environmental Considerations and Sensitive Areas:

- There are agricultural regions en route, so adherence to environmental safety standards is crucial. Avoid spillage or leakage during transport to prevent crop contamination.

9. Communication Coverage:

- Mobile network coverage is generally strong along the highway, but there may be occasional dead zones in rural stretches. Carry a GPS device and plan stops accordingly.

10. Estimated Emergency Response Times:

- Response times may vary. In urban areas, expect 15-20 minutes, while rural areas might experience delays up to 40 minutes. Have emergency contacts readily available.

11. Overall Summary of Risk Assessment:

- While the route is relatively safe with adequate infrastructure, drivers must remain vigilant during weather-related hazards, peak traffic hours, and ensure compliance with local regulations for

hazardous materials. Regular vehicle maintenance and driver training for emergency situations are advisable. Ensure comprehensive route planning is done to manage potential risks effectively.

The information here provides a basis for strategic planning and risk management for truck drivers traveling this route. Regular updates on local conditions and ongoing situational awareness will further ensure safety during transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41975, 77.88075	10 KM/Hr
5	Turn	Medium	11.41876, 77.88487	30 KM/Hr
6	Turn	Medium	11.41829, 77.88568	30 KM/Hr
7	Turn	High	11.41859, 77.88692	15 KM/Hr
8	Turn	Medium	11.41852, 77.88700	30 KM/Hr
9	Turn	Medium	11.41761, 77.88739	30 KM/Hr
10	Turn	Medium	11.41716, 77.88935	30 KM/Hr
11	Turn	Medium	11.41630, 77.89077	30 KM/Hr
12	Turn	Medium	11.41601, 77.89198	30 KM/Hr
13	Turn	Medium	11.41603, 77.89479	30 KM/Hr
14	Turn	Medium	11.41439, 77.90025	30 KM/Hr
15	Turn	Medium	11.41377, 77.90073	30 KM/Hr
16	Turn	Medium	11.41389, 77.90161	30 KM/Hr
17	Turn	Medium	11.41325, 77.90313	30 KM/Hr
18	Turn	Medium	11.41276, 77.90349	30 KM/Hr
19	Turn	Medium	11.41227, 77.90470	30 KM/Hr
20	Turn	High	11.41230, 77.90485	15 KM/Hr
21	Turn	Medium	11.41262, 77.90506	30 KM/Hr
22	Turn	Medium	11.41273, 77.90524	30 KM/Hr
23	Turn	High	11.41329, 77.90978	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
24	Turn	Medium	11.41314, 77.91006	30 KM/Hr
25	Turn	High	11.41313, 77.91057	15 KM/Hr
26	Turn	High	11.41307, 77.91058	15 KM/Hr
27	Turn	Medium	11.41334, 77.91266	30 KM/Hr
28	Turn	High	11.41424, 77.91433	15 KM/Hr
29	Turn	Medium	11.41409, 77.91458	30 KM/Hr
30	Turn	High	11.41208, 77.91551	15 KM/Hr
31	Turn	High	11.41347, 77.92096	15 KM/Hr
32	Turn	Medium	11.41336, 77.92130	30 KM/Hr
33	Turn	High	11.40566, 77.92355	15 KM/Hr
34	Turn	High	11.40557, 77.92343	15 KM/Hr
35	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
36	Turn	Medium	11.41332, 78.08072	30 KM/Hr
37	Turn	Medium	11.41463, 78.08067	30 KM/Hr
38	Turn	High	11.42964, 78.11587	15 KM/Hr
39	Blind Spot	Blind Spot	11.42631, 78.15605	10 KM/Hr
40	Turn	Medium	11.38600, 78.16120	30 KM/Hr
41	Turn	Medium	11.38570, 78.16140	30 KM/Hr
42	Turn	High	11.38557, 78.16178	15 KM/Hr
43	Turn	High	11.37866, 78.16320	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
6	hospital	Government Hospital	11.3900664, 78.1606968	30 km/h	Medium
7	police	Police station	11.3844885, 78.1622883	30 km/h	Medium
8	police	Puduchathiram Police station	11.3845086, 78.1624257	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
2	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
3	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
4	college	Gnanamani Institutions	11.4210938, 78.157164	30 km/h	Medium
5	school	Paavai Group of schools	11.3994676, 78.1594685	30 km/h	Medium
9	school	RGR CBSE School	11.3790105, 78.1615844	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41975, 77.88075



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41876, 77.88487



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41859, 77.88692



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41852, 77.88700



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41761, 77.88739



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41716, 77.88935



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41630, 77.89077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41601, 77.89198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41603, 77.89479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41439, 77.90025



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41377, 77.90073



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41389, 77.90161



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41325, 77.90313



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41227, 77.90470



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41230, 77.90485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41262, 77.90506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41273, 77.90524



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41329, 77.90978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41314, 77.91006



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41313, 77.91057



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41307, 77.91058



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41334, 77.91266



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41347, 77.92096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41336, 77.92130



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40557, 77.92343



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41332, 78.08072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.42631, 78.15605



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38600, 78.16120



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38570, 78.16140



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38557, 78.16178



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37866, 78.16320

